Switching Circuits Laboratory (CS29002)

<u>Assignment 4 (04-02-2026)</u>

a) Using the 74LS153 TTL IC (dual 4-to-1 multiplexer) and additional gates as required, implement an 8-to-1 multiplexer. Apply the inputs and observe the output through the on-board switches/LED.

Using the 8-to-1 multiplexer as designed and the minimum number of additional gates as required, implement the following 4-variable function:

$$F\ (A, B, C, D) = \Sigma\ (2, 3, 5, 8, 10, 11, 12)$$

b) There are three 4-bit data inputs $A = (A_3, A_2, A_1, A_0)$, $B = (B_3, B_2, B_1, A_0)$, and $C = (C_3, C_2, C_1, C_0)$. Based on two select lines S1 and S2, one of the three data inputs is transferred to the output $F = (F_3, F_2, F_1, F_0)$. Follow the convention:

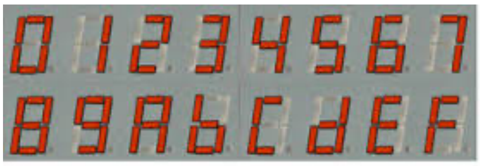
S1 = 0, S2 = 0 :: F = A

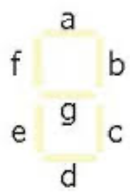
S1 = 0, S2 = 1 :: F = B

S1 = 1, S2 = X :: F = C

Implement the circuit using 74LS157 TTL IC (quad 2-to-1 multiplexer).

c) Design a 4-bit binary to hexadecimal 7-segment decoder using a 4K x 8 Erasable Programmable Read-Only Memory (EPROM) IC chip 2732. You will store the 7-bit segment codes for the 16 digits in the first 16 memory locations, and apply the data input to the four least significant address lines (with all higher-order bits set to zero). You will use an EPROM programmer to write the data bytes to the chip.





Connect a common-anode 7-segment display to the EPROM chip and verify the decoder's functionality. Connect suitable current-limiting resistor(s) to the segment-driving output lines.

<u>Instructions:</u>

1) *For each design, draw the circuit diagram on paper, following the suggested conventions. Following this, you can issue the components and start with the realization on the breadboard.*
2) *After you finish each part of the experiment, show it to your assigned TA, who will be doing the evaluation. There will be a penalty for late demonstration beyond the lab hours.*
3) *Prepare a laboratory report in PDF format and upload it to Moodle by the specified deadline. A single report has to be uploaded per group.*